

Phase 1: Data Pre-processing using Microsoft Excel

- Imported the retail sales dataset into Microsoft Excel and organized the data for preprocessing.
 - Checked the dataset for duplicate records using the **Transaction ID** column and verified that no duplicate entries existed.
 - Inspected all columns for missing values using Excel filters.
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Items

- Identified missing values in the **Item** column and replaced them with **Unknown_Item**.
 - Standardized the **Item** column by converting item names into a consistent format (Item_Number_Category), such as Item_10_PATI and Item_17_MILK.
 - Used Excel functions such as **IF()**, **TRIM()**, **MID()**, **FIND()**, **LEFT()** and **UPPER()** to automate data cleaning and standardization.
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Price Per Unit

- Checked the **Price Per Unit** column for missing (blank) values using Excel filters.
 - Calculated the **average** of the available price values using the **AVERAGE()** function.
 - Used the **IF()** and **ISBLANK()** functions to replace only the blank cells with the calculated average while keeping the existing values unchanged.
 - Applied the **Currency** number format to the column to display all price values consistently with a currency symbol and two decimal places.
 - Verified that no missing values remained in the **Price Per Unit** column.
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Quantity

- Checked the **Quantity** column for missing (blank) values using Excel filters.
 - Calculated the **median** of the available quantity values using the **MEDIAN()** function.
 - Used the **IF()** and **ISBLANK()** functions to replace only the blank cells with the calculated median while preserving the original values.
 - Verified that the **Quantity** column contained no missing values after preprocessing.
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Total Spent

- Checked the **Total Spent** column for missing (blank) values using Excel filters.
 - Calculated the **median** of the available total spent values using the **MEDIAN()** function to reduce the effect of outliers.
 - Used the **IF()** and **ISBLANK()** functions to replace only the blank cells with the calculated median while keeping the existing values unchanged.
 - Verified that no missing values remained in the **Price Per Unit** column.
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Transaction Date

- Checked the **Transaction Date** column for missing or invalid values.
 - Applied the **Short Date** format to standardize all date entries.
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Discount Applied

- Checked the **Discount Applied** column for missing (blank) values.
- Used the **IF()** and **ISBLANK()** functions to replace blank cells with **N/A**.
- Standardized the values by converting **TRUE** to **YES** and **FALSE** to **NO** for better readability and consistency.

Highlighted all imputed values for easy identification and hide the original raw columns to improve worksheet readability.

Purchase Category

- Created a **Purchase Category** calculated column based on the **Total Spent** value.
- Used the **IF()** function to classify each transaction into **High**, **Medium**, or **Low** spending categories.

- Verified the categorized values to ensure consistency and support customer purchasing pattern analysis.
 - =IF([@[Total Spent]]>=500,"High",IF([@[Total Spent]]>=200,"Medium","Low"))
 - Applied **Conditional Formatting** using **Highlight Cells Rules** to highlight **High** purchase categories with a **dark brown** font, making high-value transactions easier to identify and analyze.
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Pivot Tables

- Created Pivot Tables to summarize and analyze the cleaned dataset.
 - Generated a **Sales by Category** Pivot Table to compare the total sales across different product categories.
 - Created a **Location by Price Per Unit** Pivot Table to analyze the total price per unit based on store location (**In-store** and **Online**).
 - Developed a **Transactions by Payment Method** Pivot Table to summarize the number of transactions for each payment method.
 - Used Pivot Tables to organize the data into meaningful summaries, providing insight.
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Phase 2: Data Modeling and Visualization using Power BI Data Import

- Imported the cleaned Excel dataset into Power BI.
- Verified the data for modeling and reporting.

Data Normalization

- Created a **Retail_Sales** fact table containing all transaction records.
- Created a **Dim_Product** table by referencing the main table and retaining only the **Items** and **Category** columns.
- Created a **Dim_Customer** table by referencing the main table and retaining only the **Customer ID** column.

Data Transformation

- Removed unnecessary and hidden columns from the dimension tables.
- Removed duplicate records to maintain unique values.
- Generated unique identifiers for **Unknown_Item** values using **Index** and **Custom Column** in Power Query.

Relationship Modeling

- Established a **One-to-Many (1:*)** relationship between **Dim_Product** and **Retail_Sales** using the **Items** column.
 - Established a **One-to-Many (1:*)** relationship between **Dim_Customer** and **Retail_Sales** using the **Customer ID** column.
 - Applied **Single Cross Filter Direction** to optimize data filtering and reporting.
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Executive Dashboard Development

- Designed and developed a professional **Retail Store Sales Executive Dashboard** in Power BI to provide a high-level overview of business performance.
- Created DAX measures to calculate key business metrics, including:
 1. Total Sales
 2. Total Transactions
 3. Total Quantity Sold
 4. Average Unit Price
- Displayed the DAX measures using KPI cards for quick business insights.
- Developed interactive visualizations, including:
 1. Monthly Sales Trend (Line Chart)
 2. Sales by Category (Bar Chart)
 3. Sales by Location (Column Chart)
 4. Payment Method Distribution (Donut Chart)
 5. Discount Status Analysis (Bar Chart)

- Added interactive slicers for **Year**, **Location**, and **Category** to enable dynamic filtering.
 - Designed a professional dashboard layout with consistent colors, navigation buttons, icons, and formatting to improve readability and user experience.
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Customer Analysis Dashboard

- Created a new report page named **Customer Analysis Dashboard**.
 - Developed DAX measures:
 1. Total Customers
 2. Total Customer Sales
 3. Average Purchase per Customer
 4. Average Transactions per Customer
 - Stored all measures in the **Measures Table** for better organization.
 - Added KPI Cards to display key customer metrics.
 - Implemented interactive slicers for **Year**, **Location**, and **Category**.
 - Created the following visualizations:
 1. Top 10 Customers by Sales (Bar Chart)
 2. Purchase Category Distribution (Donut Chart)
 3. Payment Method Usage (Column Chart)
 4. Customer Sales Trend (Line Chart)
 - Maintained the same navigation panel and overall design as the Executive Dashboard for consistency.
 - Formatted KPI values by removing unnecessary decimal places and using **K (Thousands)** and **M (Millions)** notation for improved readability.
 - Applied customer-related icons and consistent styling to create a clean and professional dashboard.
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Product Analysis Dashboard Report

- Created Product Analysis Dashboard page.
 - Developed DAX measures for Total Products, Total Product Sales, Average Price per Unit, and Average Quantity Sold.
 - Added slicers for Year, Purchase Category, and Category.
 - Designed visuals to analyze purchase category distribution, quantity sold by category, monthly product sales trend, sales contribution by category, and Top 10 best-selling items.
 - Applied an Orange color theme to differentiate the dashboard from the Executive and Customer dashboards.
 - Used professional chart titles and formatted KPI values by removing unnecessary decimal places.
 - Maintained consistent navigation buttons, layout, icons, and overall dashboard design.
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Business Insights Dashboard

- Created **Business Insights Dashboard** page to provide a concise overview of key business performance and insights.
- Developed DAX measures for:
 1. Total Sales
 2. Total Transactions
 3. Discounted Transactions
 4. Average Transaction Value
- Added KPI Cards to display the key business metrics.
- Implemented interactive slicers for **Year**, **Location**, and **Category**.
- Created the following visualizations:
 1. Monthly Revenue Trend (Line Chart)
 2. Revenue by Purchase Category (Column Chart)
 3. Payment Method Distribution (Donut Chart)

4. Revenue Contribution by Category (Treemap)

- Added a **Business Insights Summary** to highlight important findings from the dashboard.
- Applied professional chart titles, consistent formatting, and suitable number formatting.
- Maintained the **Page Navigator, layout, icons, and overall dashboard design** for consistency with the other dashboards.
- Used the dashboard to identify key findings related to **revenue, purchase categories, sales trends and payment methods**.

“Created bookmarks for all five report pages to save and restore the desired dashboard state after making changes.”

Findings & Insights

1. Overall Sales Performance

- The business generated approximately **₹1.62M in total sales** across **13K transactions**.
- Approximately **70K units** were sold, with an **average price per unit of ₹23.37**.
- The **average transaction value** was approximately **₹128.64**.

2. Customer Performance

- A total of **25 customers** contributed approximately **₹1.62M in sales**.
- The **Low Purchase Category** represents the largest customer segment at **65.85%**, followed by Medium (**27.41%**) and High (**6.74%**).
- The leading customer generated sales of approximately **₹70K**, indicating a strong contribution from top customers.

3. Product Performance

- The analysis covers approximately **1K products** with total product sales of **₹1.62M**.
- The **Medium Purchase Category** generated the highest product sales contribution at **44.96%**, followed by Low (**37.09%**) and High (**17.94%**).
- Product quantities sold remain relatively consistent across the major product categories.
- The **Top 10 Best-Selling Items** identify the products contributing the highest individual sales.

4. Business Insights

- **Medium Purchase Category** is the leading revenue-generating purchase segment.
- **Online sales slightly exceed In-store sales**, making online sales the stronger sales channel.
- **Cash, Digital Wallet, and Credit Card** usage is relatively balanced, indicating diversified payment preferences.
- Sales and revenue show a **significant January peak**, followed by a comparatively stable pattern throughout the remaining months.
- The business recorded approximately **4K discounted transactions**, highlighting substantial use of promotional pricing.

Overall Insight

The business demonstrates strong sales performance with ₹1.62M in revenue. Medium Purchase Category leads product revenue, while Low Purchase Category represents the largest customer segment. Online sales slightly outperform In-store sales, payment methods remain well balanced, and monthly performance stabilizes after the January peak.